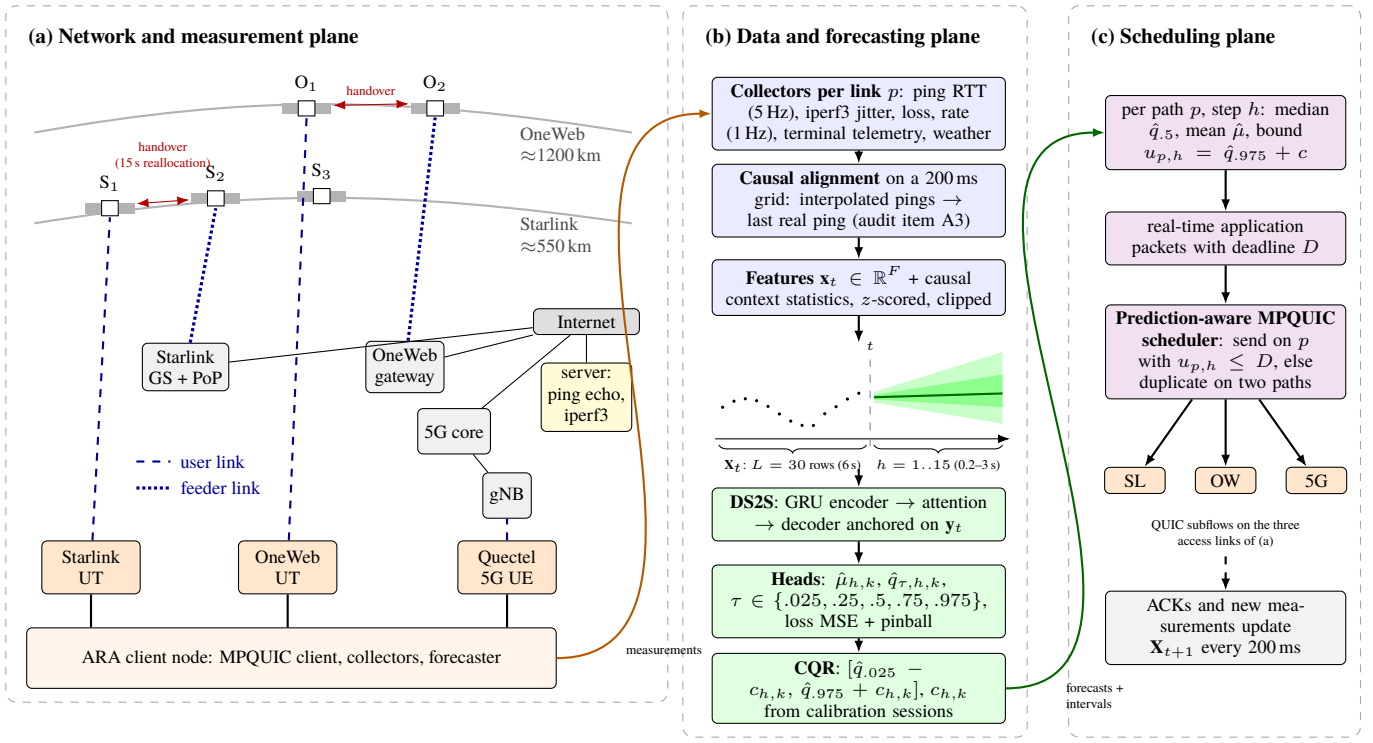
(a) Network and measurement plane
O1
handover
O2
OneWeb
≈1200 km
handover
(15 s reallocation)
S1
S2
S3
Starlink
≈550 km
Internet
Starlink
GS + PoP
OneWeb
gateway
server:
ping echo,
iperf3
5G core
gNB
user link
feeder link
Starlink
UT
OneWeb
UT
Quectel
5G UE
ARA client node: MPQUIC client, collectors, forecaster
measurements
(b) Data and forecasting plane
Collectors per link p: ping RTT
(5 Hz), iperf3 jitter, loss, rate
(1 Hz), terminal telemetry, weather
Causal alignment on a 200 ms
grid: interpolated pings →
last real ping (audit item A3)
Features x_t ∈ ℝ^F + causal
context statistics, z-scored, clipped
t
x_t: L = 30 rows (6 s) h = 1..15 (0.2–3 s)
DS2S: GRU encoder → attention
→ decoder anchored on y_t
Heads: μ̂_{h,k}, q̂_{τ,h,k},
τ ∈ {.025, .25, .5, .75, .975},
loss MSE + pinball
CQR: [q̂.025 −
c_{h,k}, q̂.975 + c_{h,k}], c_{h,k}
from calibration sessions
forecasts +
intervals
(c) Scheduling plane
per path p, step h: median
q̂.5, mean μ̂, bound
u_{p,h} = q̂.975 + c
real-time application
packets with deadline D
Prediction-aware MPQUIC
scheduler: send on p
with u_{p,h} ≤ D, else
duplicate on two paths
SL
OW
5G
QUIC subflows on the three
access links of (a)
ACKs and new mea-
surements update
X_{t+1} every 200 ms